Md. Mahfuz Haque

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Dept. of Local Government & Urban Development, JKKNIU, Trishal, Mymensingh 2224, Bangladesh
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EDUCATION

Jatiya Kabi Kazi Nazrul Islam University (JKKNIU) *Dec 2023 – Dec 2026*

Bachelor of Social Science in Local Government and Urban Development — 4th Year, 2nd Semester

- **Thesis**: Understanding Travel-Behavior Change in Response to Urban Waterlogging in Bangladesh: An Integrated GIS–SEM–Machine Learning Study of Mymensingh, with Dhaka as a Comparative Case.
- **Relevant Coursework**: Urban Planning & Development; Environmental Design; Public Health; Local Governance, Law and Policy
- Cumulative GPA: 3.51/4.00 (Current) | **Honors**: University (Central) Scholarship (2024, 2025)

RESEARCH INTERESTS

Urban & Regional Planning | Sustainable Urban Development | GIS & Remote Sensing | Transport & Mobility | Environmental Design & Crime Prevention | Public Health & Urban Inequality | Local Governance & Policy | Geospatial Data Science

RESEARCH & PROJECTS

PEER-REVIEWED JOURNAL ARTICLES

1. Hossain, M. I., Mazhar, M., ***Haque, M. M.***, Hossain, H., Rahman, M. M., & Rahman, A. (2026). ***Service Quality, User Heterogeneity, and Policy Priorities in Suburban Paratransit: A Case Study from Trishal, Bangladesh***. *Case Studies on Transport Policy* (ELSEVIER), DOI: 10.1016/j.cstp.2026.101998

- Surveyed 104 suburban paratransit users on 28 service-quality attributes; applied sequential EFA–CFA (R, Python) to identify a three-dimensional second-order service-quality construct, Mann–Whitney U tests to assess gender and income heterogeneity, and importance–performance analysis (IPA) to generate a policy prioritisation framework.

MANUSCRIPTS IN PREPARATION

1. **Trends in Environmental Design and Crime Deterrence: A GIS-Based Analysis of Mymensingh** *Ongoing*

A GIS-Based Analysis of Mymensingh

- Applying GIS tools to analyze decade-long relationships between urban environmental design and crime deterrence patterns in Mymensingh.

2. **Spatiotemporal Dynamics and Predictive Modeling of Traffic Accidents Across the Bangladesh Road Hierarchy** *Ongoing*

Integrating GIS, Machine Learning, and Multi-Source 15-Year Accident Data (National, Regional, Zila, and Local Roads)

- Developing a robust predictive geospatial framework utilizing a triangulated 1year dataset from police records, media reports, and infrastructure databases to identify persistent accident hotspots and forecast crash severity across the complete Bangladesh road network using Machine Learning (Random Forest, XGBoost, CNN-LSTM) and GIS-based spatiotemporal analysis.

3. **A Comparative Analysis of Clinical Waste Management Systems: A Mixed-Methods and Machine Learning Approach** *Ongoing*

Cox's Bazar Medical College (Coastal) vs. Mymensingh Medical College (Inland), Bangladesh

- Comparing clinical waste disposal risk profiles across coastal and inland tertiary hospitals using Leachate Pollution Index (LPI), Environmental Risk Assessment (ERA/Hazard Index), and a constructed Waste Management Efficiency Index

(WMEI), integrating GIS spatial analysis and Machine Learning to assess how geographic location modulates environmental and public health outcomes.

RESEARCH EXPERIENCE

1. **Sexual Harassment in Higher Educational Institutions in Bangladesh: A Study at Jatiya Kabi Kazi Nazrul Islam University** *FY 2024-25*

Undergraduate Research Assistant, Data Analyst | Director of Research and Extension (R&E)

- Led a multi-disciplinary research team to assess insights and performed qualitative and quantitative analysis under the supervision of the Chairman and Assistant Professor, Department of Local Government and Urban Development.

2. **Consequences of Studying Unwanted Subjects at Public University in Bangladesh** *Ongoing*

Undergraduate Research Assistant |UGC-Funded Research Project, University Grants Commission (UGC), Bangladesh

- Examining socio-academic consequences faced by students enrolled in non-preferred disciplines at public universities.

3. **Center for Bio-Health Research Bangladesh** *Ongoing*

Research Collaborator — Supervised by Prof. Dr. Zakir Masud (Post-doctoral Fellow, Univ. of Alabama at Birmingham, USA)

- Contributing to interdisciplinary public health research under the advisory of Dr. Saif Mahabub (PhD Fellow, Univ. of Cincinnati, USA) and Prof. Dr. Nazmul Huda, Dept. of Ayurvedic Medicine, Dhaka Indigenous Medical College and Hospital, Mirpur-13, Dhaka.

PROFESSIONAL EXPERIENCE

Dept. of Local Government and Urban Development, JKKNIU — Teaching Assistant *Oct 2024 – Present*

- Conducted instructional classes for undergraduate students on advanced research methodologies.
- Taught practical applications of qualitative and quantitative data analysis using NVivo and SPSS.
- Instructed students in GIS tools for urban planning and spatial data mapping.

Dept. of Local Government and Urban Development, JKKNIU — Research Assistant *Jul 2025 – Present*

- Assisted faculty with comprehensive research projects, supporting both qualitative and quantitative analysis.
- Managed primary data collection and organization processes to ensure accuracy and integrity of research findings.
- Collaborated on research papers, conference papers, reports, and formal research proposals.

Private Tutor — O Level, A Level & University Admission *2018 – Present*

- Tutored O Level and A Level students across core Science and Humanities subjects.
- Prepared public university admission candidates who successfully gained admission to top institutions under personal guidance.
- Designed personalized study plans and mock assessment frameworks improving student performance significantly.

Urban Kitchen and Bath (UrbanKitchenAndBath.ca) — Design Drafter & Qualitative Analyst *Remote | Ongoing*

Toronto's Kitchen Refacing Specialist

- Prepared draft floor plans and design layouts for kitchen refacing and renovation projects.
- Conducted qualitative analysis of customer preferences and requirements to inform design decisions and improve client satisfaction.

LEADERSHIP EXPERIENCE

Dept. of Local Government and Urban Development, JKKNIU — Main Host & Programme Coordinator, LGUD Bloom Fest *Feb 2026*

- Served as Main Host and full Programme Coordinator for LGUD Bloom Fest 2026, the department's flagship cultural event, engaging 4,000+ attendees including participants from other departments and the wider local community.
- Coordinated end-to-end event logistics, stage management, and audience engagement, ensuring smooth execution of the full-day programme.

Skill Development Club (SDC), JKKNIU — Executive, Public Relations Team *FY 2024/25*

- Facilitated key networking initiatives and served as primary spokesperson for organizational events.
- Spearheaded outreach efforts and secured sponsorships to fund major club events.

Dept. of Local Government and Urban Development, JKKNIU — Class Representative *2024*

Session 2021–22

- Represented the student cohort in all university and departmental programs as the primary liaison.
- Managed academic logistics, organized class schedules, and coordinated midterm examination dates with faculty.
- Served as management head for departmental events, overseeing planning, logistics, and execution.

Cantonment Public School and College Momenshahi — Class Captain & Football Team Organizer *Jul 2018 – Dec 2019*

- Liaised between faculty and students to maintain smooth administrative communication.
- Served on the organizing committee for the college football team, coordinating logistics and team events.

HONORS AND AWARDS

Ministry of Housing and Public Works, Govt. of Bangladesh — National Finalist *2025*

- Selected as a finalist in "Youth Perspectives on Sustainable Urban Development" organized by the Urban Development Directorate (UDD).

Jatiya Kabi Kazi Nazrul Islam University — Project Presenter *2025*

- Presented an independent research project at the 3rd Research Fair – 2025.

Janata Bank — Outstanding Student Award & Scholarship *2018*

- Awarded a merit-based scholarship of BDT 20,000 for exceptional academic performance (GPA 5.00) in SSC.

Mymensingh City Corporation — Mayor's Outstanding Student Medal *2018*

- Honored with a prestigious medal by the City Mayor for academic excellence.

SKILLS AND INTERESTS

Language

- Native Bengali; Fluent English

Computer Skills

Geospatial & Remote Sensing (GIS):

- ***GIS Ecosystem:*** ArcGIS Pro, ArcMap, QGIS, Google Earth Engine (with AI integrations), Google Earth Pro, Google My Maps
- ***Spatial Analysis:*** Geoprocessing, Interpolation, Zonal Statistics, Location Analysis, Stream Order, Terrain Analysis (Slope, Hillshade, Contour), Elevation Measurement
- ***Remote Sensing:*** Satellite Image Processing, USGS Earth Explorer, DEM Data, Landsat 8 Band Composition, Raster Calculator (NDVI, NDWI, NSMI), Supervised/Unsupervised Classification
- ***Mapping & Digitizing:*** Georeferencing, Feature & Mouza Map Digitizing, Attribute Table Operations, Map Layout & Symbology, Demographic Mapping, Flood Risk Zonation

Programming & Version Control:

- ***Python (Core & OOP):*** Core Python & OOP: Data Structures, Functions, File & Exception Handling, Classes, Inheritance, Polymorphism, Decorators, Design Patterns
- ***SQL & Databases:*** RDBMS, Database Normalization, Advanced Queries, Joins, Set Operations
- ***Git & GitHub:*** Repository Management, Branching & Merging, Conflict Resolution, Remote Repositories, Pull Requests, Version History

Machine Learning & Artificial Intelligence:

- ***Core Concepts:*** Supervised, Unsupervised & Reinforcement Learning; Train/Test/Validation Splits; Model Optimization
- ***Algorithms:*** Linear & Logistic Regression, KNN, Decision Tree, SVM, K-Means Clustering, Naive Bayesian Classifier
- ***Data Preprocessing & Evaluation:*** Feature Extraction, Encoding & Scaling, Outlier Detection, Confusion Matrix, Precision, Recall, F1 Score
- ***Advanced AI & NLP:*** Natural Language Processing Pipeline, NLTK, Ambiguity Resolution, Neural Networks

Data Science, Statistics & Visualization:

- ***Data Manipulation:*** NumPy (Arrays, Indexing, Slicing), Pandas (DataFrames, CSV/Excel Handling, Aggregation, Data Cleaning)
- ***Statistics (EDA):*** Central Tendency, Dispersion, Normal Distribution, Coefficient of Correlation
- ***Visualization:*** Matplotlib, Seaborn, Plotly, Time Series/Network/Text Data Visualization, Dashboard Design & Development

Research Methodologies & Tools:

- SPSS, NVivo, KoBo Toolbox, Zotero, Qualitative & Quantitative Data Analysis, Literature Review, Research Proposal Writing, Referencing Standards

Online Courses & Continuing Education

- Complete ArcGIS Pro Mastery: A Hands-On Practical Course — Udemy
- GIS Certification Program — BetterSkill.io
- Data Science & Machine Learning with Python — Ostad App